# Endometriosis AI Analysis Report

*Advanced Multi-Modal Federated Prediction*

## 1. Patient Clinical Metrics

| Age | BMI | Pelvic Pain | Dysmenorrhea |
|---|---|---|---|
| 35 | 26.8 | 7/10 | 7/10 |

| Dyspareunia | Family Hist. | CA-125 | Estradiol |
|---|---|---|---|
| Yes | Yes | 54.2 U/mL | 220.0 pg/mL |

| Progest. |
|---|
| 6.1 ng/mL |

## 2. AI Diagnostic Prognosis

**Confidence Score: 0.0% (+/- 4.0%)**

**Estimated Stage: Stage II (Mild)**

**Future Risk Trajectory (Years 1, 3, 5):**
1-Year: 43.6%,  3-Year: 54.2%,  5-Year: 47.5%

### Generative Clinical Synthesis:

DISCLAIMER: AI research tool  not a clinical diagnosis.

Primary Assessment: 0.0% (+/-4.0%) probability of active endometriosis. Estimated classification: Stage II (Mild).

Neural Routing: 53.2% of computation routed to the Adenomyosis/Uterine Expert Sub-Network.

Biomarker Highlights:
- CA-125: 54.2 U/mL  elevated (threshold 35 U/mL). Differential diagnosis required.
- Estradiol: 220.0 pg/mL  hyperestrogenism may drive ectopic proliferation.
- Pelvic Pain Score 7/10  indicates significant nociceptive involvement.

Recommendation: High-resolution transvaginal ultrasound and pelvic MRI for structural assessment.

## Personal Actionable Plan:

Personal Actionable Health Plan (AI-generated for educational purposes only):

1. Monitoring: Risk profile is stable. Maintain routine gynaecological checkups.
2. Pain Management: Pain score is elevated  discuss NSAIDs or pelvic floor physiotherapy.
3. Hormonal Balance: Omega-3-rich anti-inflammatory diet may help regulate excess oestrogen.
4. Lifestyle: An anti-inflammatory diet (Mediterranean) can reduce systemic inflammation markers.

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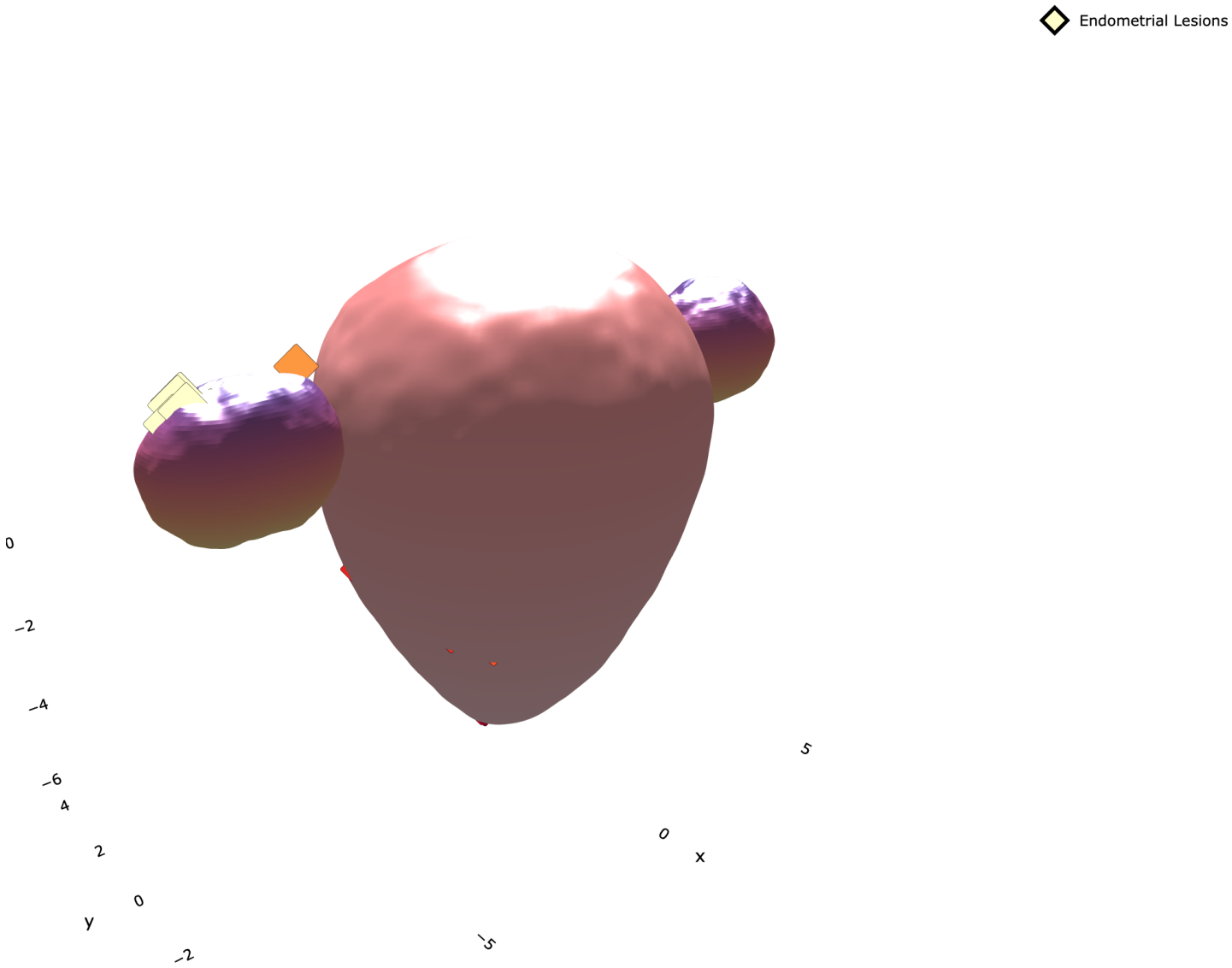
## 3. Physics-Informed Digital Twin (3D Render)

High-resolution rendering of the generated patient organ geometry, mapping lesions, endometriomas, and simulated adhesions based on the PINN predictions.



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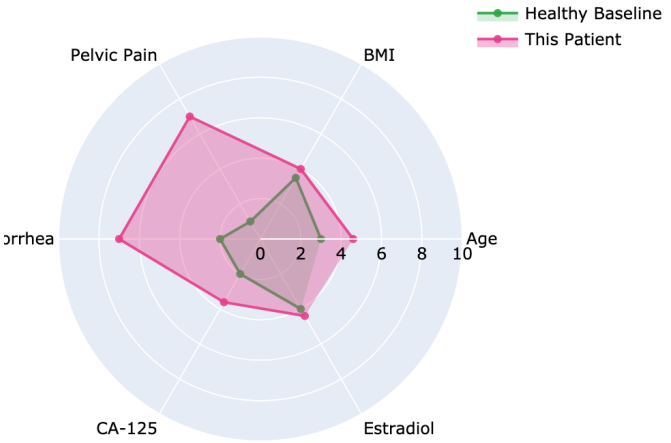
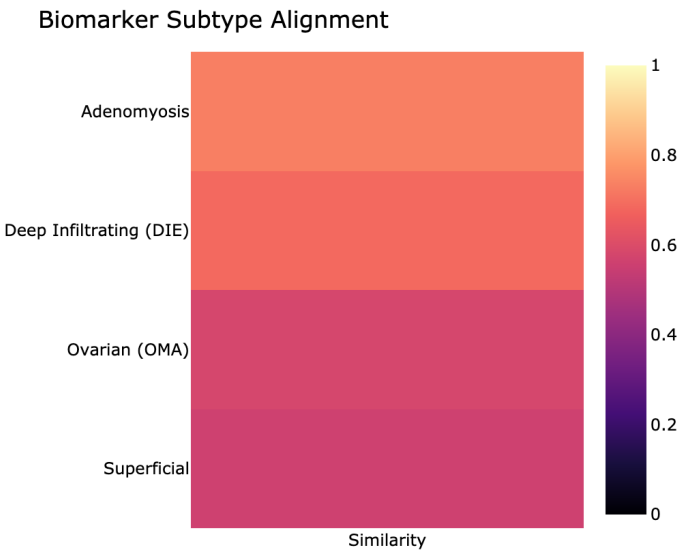
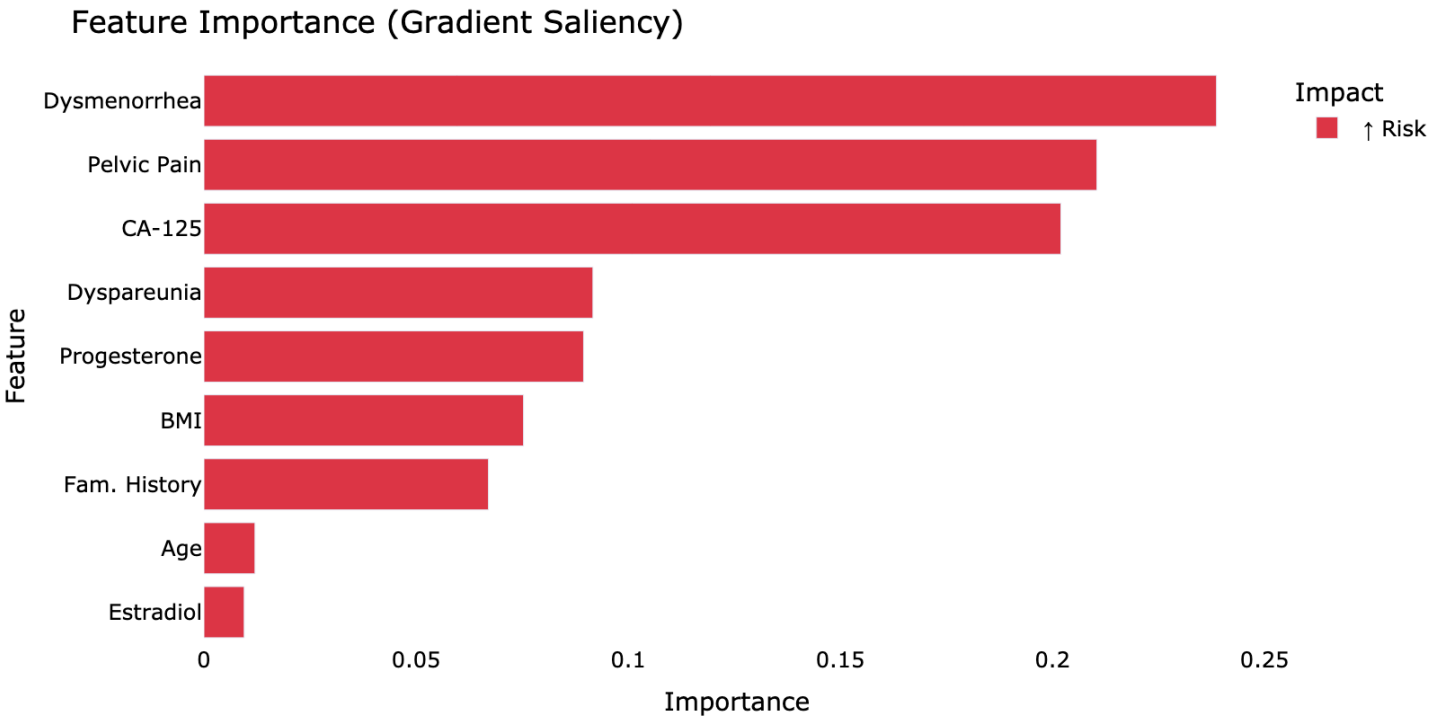
## 4. Advanced Analytics & Explainable AI




### Feature Importance (SHAP/XAI Explainer):




XAI Insight: Risk score 0.0%. Primary drivers: Pelvic Pain (7.0), Dysmenorrhea (7.0).